The bridge midpoint on a compact group at any cut: windings and the centre

Let G be a compact connected simply connected Lie group with a bi-invariant metric, let $t > 0$, and take the heat-kernel bridge of total heat time t from e to $g = e^{iX}$, cut at fraction $s \in (0,1)$, so that the first piece has heat time st. For every irreducible representation λ the midpoint expectation of its character is an exact finite sum, over the weights κ of λ, of image sums over the coroot lattice $Q^\vee$ (Theorem 1):

$$
\begin{aligned}
E\,\chi_\lambda(m) &= D_t(X)^{-1} \sum_\kappa m_\lambda(\kappa) e^{-ts(1-s)|\kappa|^2/2} e^{is\langle \kappa, X\rangle} I_{(1-s)\kappa}(X), \\
I_c(X) &= \sum_{H \in Q^\vee} e^{2\pi i \langle c, H\rangle} \pi\Big(\tfrac{i}{t}(X - 2\pi H) - c\Big) G_H, \qquad (1) \\
D_t(X) &= I_0(X), \qquad G_H = e^{-|X-2\pi H|^2/(2t)}.
\end{aligned}
$$

with $m_\lambda(\kappa)$ the weight multiplicities and $\pi(v) = \prod_{\alpha>0}\langle v, \alpha\rangle/\langle \rho, \alpha\rangle$ the Weyl dimension polynomial.

- **Corollary 2 (reductions).** For $U(1)$, (1) is the winding mixture of Corollary 5_s(a). For $SU(2)$ at $s = \frac{1}{2}$ it is Theorems 1 and 2 of the $SU(2)$ midpoint note, image terms included.
- **Corollary 3 (the centre).** The phase $e^{2\pi i(1-s)\langle \kappa, H\rangle}$ of an image is the same for all weights of every representation iff $(1-s)H \in P^\vee$, the coweight lattice, and it is then the central character of λ at $z_H = e^{2\pi i(1-s)H}$. At $s = p/q$ in lowest terms these central images reach exactly the q-torsion of the centre: $\mathbb{Z}_{\gcd(q,N)}$ for $SU(N)$.
- **Corollary 4 ($SU(3)$).** At $s = \frac{1}{3}$ the central images are $H \in 3P^\vee$; the smallest are the two Weyl triples of $\pm(1,1,-2)$, of relative weight $e^{-24\pi^2/t}$ at $X = 0$ before polynomial prefactors and the regular limit, and they carry ω^2 and ω on the fundamental ($\omega = e^{2\pi i/3}$), the conjugates on the antifundamental and 1 on the adjoint. At $s = \frac{1}{2}$ every image with a nontrivial phase has a weight-dependent one: a halving reaches only the identity in the centre.

For $SU(2)$ at $s = \frac{1}{2}$ the phase is $(-1)^w$ for half-integer spin and 1 for integer spin. This is Dirac's belt trick, $\pi_1(SO(3)) = \mathbb{Z}_2$ (Newman 1942, metadata), appearing algebraically in the exact midpoint law: the Cartan image label w shifts a torus geodesic by $4\pi w$, and its midpoint by $2\pi w$, which spinor characters detect. Since $\pi_1(SU(2)) = 0$, these labels describe relative torus geodesics, rather than homotopy sectors of paths in the group. The spatial belt trick is a comparison through $SU(2) \to SO(3)$. In specific monopole backgrounds, internal and spatial rotations also meet in "spin from isospin" (Jackiw and Rebbi 1976; Hasenfratz and 't Hooft 1976; metadata). For $SU(3)$ the same mechanism carries the triality, and only cuts at thirds (or at denominators divisible by 3) produce it.

The ingredients are character orthogonality, the Brauer–Klimyk rule and Poisson summation, as in the $SU(2)$ note; the denominator of (1) is the classical image formula for the heat kernel on a compact group (Fegan 1983, metadata). The same unfolding and coroot Poisson summation give the heat trace at the identity and its exponential Seeley–DeWitt structure, $a_k = (R/6)^k/k!$, in the author's earlier project (physres1, Remark D9.1o″, 2026-02, read); at $X \to 0$ the $H = 0$ term of the denominator of (1) gives $k_t(e) \propto t^{-\dim G/2} e^{t|\rho|^2/2}$, the same structure ($R/6 = |\rho|^2$ in this metric; for $SU(2)$, radius 2, $R/6 = \frac{1}{4}$). This describes the local asymptotic sector; nonzero images give exponentially small corrections at the identity. With generator $\Delta/2$, the coefficients in powers of t are $(R/12)^k/k!$; $(R/6)^k/k!$ uses heat time for Δ. No novelty is claimed for the method.

The Cartan-image amplitudes are signed polynomials. A positive winding mixture is available for $U(1)$, while the simply connected nonabelian image labels are not topological winding

sectors. Thin-vortex placement remains a semiclassical interpretation. Checks are internal written derivations.

1. Conventions

The group metric is that of the series/parallel note §1: $-\Delta_G\chi_\lambda = C_2(\lambda)\chi_\lambda$ with $C_2(\lambda) = |\lambda+\rho|^2 - |\rho|^2$ in the inner product on weights dual to the metric on the Cartan algebra $\mathfrak{t}$. For $SU(N)$ roots have $|\alpha|^2 = 1$ and coroots $\alpha^\vee = 2\alpha/|\alpha|^2$ have $|\alpha^\vee|^2 = 4$, which is the metric $|Y|^2 = 2\,\mathrm{tr}\,Y^2$ on $\mathfrak{t}$; then $C_2(j) = j(j+1)$ for $SU(2)$ and $C_2(\mathbf{3}) = \frac{4}{3}$ for $SU(3)$. Torus elements are e^{iX}, $X \in \mathfrak{t}$; the pairing of the weight lattice P with the coroot lattice $Q^\vee$ is integral, P and $Q^\vee$ are dual lattices, and $e^{iX} = e$ iff $X \in 2\pi Q^\vee$ (G simply connected). The coweight lattice $P^\vee$ is dual to the root lattice Q, and $P^\vee/Q^\vee \cong Z(G)$ through $Y \mapsto e^{2\pi i Y}$. For $SU(2)$, $|X| = \theta$ is the rotation angle of the $SU(2)$ note, and $X - 2\pi w\alpha^\vee$ has length $|\theta - 4\pi w|$.

Here iX/t in the argument of π means $iX^\flat/t$, using the metric to identify $\mathfrak{t}$ with $\mathfrak{t}^*$. In diagonal $SU(N)$ coordinates, covectors a have norm $|a|^2 = \frac{1}{2}\sum a_i^2$, Cartan vectors H have norm $|H|^2 = 2\sum H_i^2$, and their pairing is $\sum a_i H_i$. Thus a root $e_i - e_j$ has squared norm 1, its coroot is the diagonal vector with entries $1, -1$, of squared norm 4, and $X^\flat = 2X$. Pairings between two Cartan vectors below use $2\sum X_i H_i$.

Heat kernel $k_t = \sum_\lambda d_\lambda e^{-tC_2(\lambda)/2}\chi_\lambda$. The bridge midpoint m has Haar density

$$k_{st}(m)\, k_{(1-s)t}(m^{-1}g)/k_t(g).$$

the bridge law (4) of the series/parallel note becomes this one under $y = P_e m^{-1}$, $g = P_e Q_e$ (with the two heat times assigned as there), and Corollary 2_s there gives the cut faces heat times st and $(1-s)t$.

Weyl: $A_v(X) = \sum_{w\in W}\varepsilon(w)e^{i\langle wv, X\rangle}$, $\chi_\lambda = A_{\lambda+\rho}/A_\rho$, $d_\lambda = \pi(\lambda+\rho)$. Extend $\chi_\nu := A_{\nu+\rho}/A_\rho$ to every $\nu \in P$: it is zero or $\pm$ an irreducible character, and $C_2(\nu) := |\nu+\rho|^2 - |\rho|^2$ is invariant under the shifted Weyl action, so it is the Casimir of that character.

Brauer–Klimyk rule. $\chi_\lambda\chi_\mu = \sum_\kappa m_\lambda(\kappa)\chi_{\mu+\kappa}$ for dominant μ. Indeed $\chi_\lambda A_{\mu+\rho} = \sum_\kappa m_\lambda(\kappa)e^{i\langle\kappa,X\rangle}\sum_w \varepsilon(w)e^{i\langle w(\mu+\rho),X\rangle}$, and replacing κ by $w\kappa$, which leaves m_λ invariant, gives $\sum_\kappa m_\lambda(\kappa)A_{\mu+\kappa+\rho}$.

2. Theorem 1 and its proof

Theorem 1 (A1: ACCEPT, conventions refined). For $A_\rho(X) \neq 0$, (1) holds; on the affine Weyl walls it means the continuous limit of the complete quotient. Individual image ratios can be singular there.

Proof. (i) *Characters.* Expanding both heat kernels gives

$$k_t(g)E\chi_\lambda(m) = \sum_{\mu,\nu} d_\mu d_\nu e^{-stC_2(\mu)/2-(1-s)tC_2(\nu)/2} \times \int \chi_\lambda(m)\chi_\mu(m)\chi_\nu(m^{-1}g)\,dm.$$

The Brauer–Klimyk rule and $\int \chi_a(m)\chi_b(m^{-1}g)dm = \delta_{ab}\chi_a(g)/d_a$ give

$$k_t(g)\,E\chi_\lambda(m) = \sum_{\mu\ \text{dominant}} d_\mu \sum_\kappa m_\lambda(\kappa)\, e^{-stC_2(\mu)/2-(1-s)tC_2(\mu+\kappa)/2}\,\chi_{\mu+\kappa}(g).$$

(ii) *Unfolding.* Multiply by $A_\rho(X)$ and put $v = \mu+\rho$, which runs over the strictly dominant weights. The summand becomes $\pi(v)\,m_\lambda(\kappa)\,e^{-\frac{t}{2}[s|v|^2+(1-s)|v+\kappa|^2-|\rho|^2]}A_{v+\kappa}(X)$. Expand $A_{v+\kappa}$ and substitute $v \mapsto w^{-1}v$, $\kappa \mapsto w^{-1}\kappa$: $\varepsilon(w)\pi(w^{-1}v) = \pi(v)$, and m_λ and the norms are Weyl invariant. Each regular weight is the image of exactly one strictly dominant weight, and π vanishes on the walls, so

$$A_\rho(X)k_t(g)E\chi_\lambda(m) = e^{t|\rho|^2/2}\sum_\kappa m_\lambda(\kappa)\sum_{v\in P}\pi(v)\,e^{-\frac{t}{2}[s|v|^2+(1-s)|v+\kappa|^2]}e^{i\langle v+\kappa,X\rangle}.$$

More explicitly, set $v' = wv$, $\kappa' = w\kappa$ in each Weyl term. Then $\varepsilon(w)\pi(v) = \pi(v')$. Strictly dominant integral weights are precisely $\rho + P_+$, since their simple-coroot coordinates are positive integers. The disjoint open chambers therefore exhaust the regular part of P exactly once, with no factor $|W|$. The wall terms added to this unfolded sum vanish through $\pi(v')$, even when $v' + \kappa'$ lies off a wall.

(iii) *Square.* $s|v|^2 + (1-s)|v+\kappa|^2 = |v + (1-s)\kappa|^2 + s(1-s)|\kappa|^2$.

(iv) *Poisson summation* over P, whose dual lattice is $Q^\vee$:

$$\sum_{v\in P} f(v) = \mathrm{vol}(\mathfrak{t}^*/P)^{-1}\sum_{H\in Q^\vee}\hat f(H).$$

Here $\hat f(H) = \int f(y)e^{-2\pi i\langle y,H\rangle}dy$. With $y = u-(1-s)\kappa$ one has $\langle y+\kappa, X\rangle = \langle u, X\rangle + s\langle\kappa, X\rangle$ and $\langle y, H\rangle = \langle u, H\rangle - (1-s)\langle\kappa, H\rangle$, so

$$\hat f(H) = e^{is\langle\kappa,X\rangle}e^{2\pi i(1-s)\langle\kappa,H\rangle}\int \pi(u-(1-s)\kappa)\,e^{-t|u|^2/2}e^{i\langle u,Z\rangle}du, \qquad Z = X - 2\pi H.$$

(v) *The Gaussian integral.* π is harmonic: $\Delta\pi$ is Weyl-antisymmetric of lower degree, and every Weyl-antisymmetric polynomial is divisible by π. Its derivatives are harmonic too. For a harmonic homogeneous h of degree d, Hobson's formula $h(\partial)F(|Z|^2) = 2^d h(Z)F^{(d)}(|Z|^2)$ gives $h(-i\partial_Z)e^{-|Z|^2/(2t)} = h(iZ/t)\,e^{-|Z|^2/(2t)}$. Taylor-expanding $\pi(u-c)$ in c into harmonic pieces and summing back,

$$\int \pi(u-c)\,e^{-t|u|^2/2+i\langle u,Z\rangle}du = \Big(\frac{2\pi}{t}\Big)^{n/2}\pi\Big(\frac{iZ}{t} - c\Big)e^{-|Z|^2/(2t)}, \qquad n = \dim\mathfrak{t}.$$

(vi) The trivial representation ($\kappa = 0$ only) gives the denominator. The constants $e^{t|\rho|^2/2}$, $\mathrm{vol}(\mathfrak{t}^*/P)^{-1}$ and $(2\pi/t)^{n/2}$ cancel. All sums converge absolutely. □

Theorem 1′ (several cuts; A1': ACCEPT). Cut the same bridge at $0 < s_1 < \cdots < s_n < 1$, with layer values $m_1, \dots, m_n$, and take irreducible representations $\lambda_1, \dots, \lambda_n$. With $\boldsymbol\kappa = (\kappa_1, \dots, \kappa_n)$ running over weights of $\lambda_1, \dots, \lambda_n$, $A(\boldsymbol\kappa) = \sum_j (1-s_j)\kappa_j$ and the bridge covariance form $Q(\boldsymbol\kappa) = \sum_{i,j}\min(s_i, s_j)\,(1 - \max(s_i, s_j))\langle\kappa_i, \kappa_j\rangle$,

$$\begin{aligned} E\prod_j \chi_{\lambda_j}(m_j) = D_t(X)^{-1}\sum_{\boldsymbol\kappa}\prod_j m_{\lambda_j}(\kappa_j) \\ \times\, e^{-tQ(\boldsymbol\kappa)/2}e^{i\langle\sum_j s_j\kappa_j, X\rangle}I_{A(\boldsymbol\kappa)}(X). \end{aligned} \tag{1′}$$

Proof. Integrate $m_1, \dots, m_n$ in turn. Each step applies the Brauer–Klimyk rule, which holds with the extended characters χ_ν for every $\nu \in P$ (its proof uses only the Weyl invariance of the weight multiset), and orthogonality; this gives $k_t(g)E\prod\chi_{\lambda_j}(m_j) = \sum_\mu d_\mu \sum_{\boldsymbol\kappa}\prod m(\kappa_j)e^{-\frac{t}{2}[s_1C_2(\mu) + \sum_j (s_{j+1}-s_j)C_2(\mu+\kappa_1+\cdots+\kappa_j)]}\chi_{\mu+\sum\kappa_j}(g)$, $s_{n+1} = 1$. The unfolding of step (ii) holds because the multiset of tuples $\boldsymbol\kappa$ is invariant under the simultaneous Weyl action. Completing the square, $s_1|v|^2 + \sum_j (s_{j+1}-s_j)|v+\kappa_1+\cdots+\kappa_j|^2 = |v+A|^2 + Q$, where Q is the covariance of the standard Brownian bridge at the times s_j, paired with the κ_j. Steps (iv)–(vi) are unchanged with $(1-s)\kappa$ replaced by A and $s\kappa$ by $\sum_j s_j\kappa_j$. □

The iteration can be justified without dividing by dimensions of extended characters. Heat convolution sends every extended χ_ν to $e^{-uC_2(\nu)/2}\chi_\nu$; a wall character stays zero. Multiplication by the next character obeys Brauer–Klimyk even on a wall, where its signed terms cancel. Heat convolution preserves that zero sum because terms representing the same signed irreducible have the same Casimir. This proves the iterated formula before unfolding. For the square, put $B(u) = \sum_{s_j \le u} \kappa_j$. Then $\int_0^1 B(u)du = A$ and $\int_0^1 |B(u)|^2 du - |A|^2 = \sum_{i,j}[1 - \max(s_i, s_j) - (1 - s_i)(1 - s_j)]\langle\kappa_i, \kappa_j\rangle = Q$. This also fixes every cross-term and sign in $(1')$.

The polynomial ratio for a single image is $\pi(\frac{i}{t}Z - (1-s)\kappa)/\pi(\frac{i}{t}Z) = \prod_{\alpha>0}(1 + i(1-s)t\langle\kappa,\alpha\rangle/\langle Z,\alpha\rangle)$, the curvature correction; for $SU(2)$ at $s = \frac{1}{2}$ it is the factor $1 + ikt/(2\theta)$ behind the $-\frac{t}{2\theta}k\sin\frac{k\theta}{2}$ term of the $SU(2)$ note.

3. Reductions (Corollary 2)

$U(1)$. The abelian calculation uses $P = Q^\vee = \mathbb{Z}$, $\pi \equiv 1$, $\rho = 0$, $\langle n, H\rangle = nH$, $X = \theta$:

$$E\, e^{inm} = e^{-ts(1-s)n^2/2}\frac{\sum_H e^{in(s\theta + 2\pi(1-s)H)}G_H}{\sum_H G_H}.$$

This is the characteristic function of the mixture over the winding H, with weights $\propto e^{-(\theta-2\pi H)^2/(2t)}$, of wrapped Gaussians of variance $s(1-s)t$ centred at $s\theta + 2\pi(1-s)H$. This is Corollary 5_s(a) of the series/parallel note.

To match that note literally, use $y = P_e - m$, $\theta = \phi_e = P_e + Q_e$ and $H = -W$. Transforming back gives $m = P_e - s\phi_e + 2\pi sH \equiv \bar{m}_e + 2\pi(1-s)W$ modulo 2π, and the image weight is $e^{-(\phi_e + 2\pi W)^2/(2t_e)}$.

$SU(2)$, $s = \frac{1}{2}$. The weights of spin J are $\kappa = k\alpha$, $k = -J, \dots, J$, with $|\kappa|^2 = k^2$, $\langle\kappa, X\rangle = k\theta$, $\langle\kappa, w\alpha^\vee\rangle = 2kw$, and $\pi(v) = 2\langle v, \alpha\rangle$. The phase is $e^{2\pi ikw}$: $(-1)^w$ for half-integer J and 1 for integer J. With $\Xi_{(\sigma)}, \Theta_{(\sigma)}$ as in the $SU(2)$ note, the numerator of (1) is $\sum_k e^{-tk^2/8}e^{ik\theta/2}\big[\frac{i}{t}\Xi_{(\sigma)} - \frac{k}{2}\Theta_{(\sigma)}\big]$ and the denominator is $\frac{i}{t}\Xi_+$. These numerator and denominator expressions have both been divided by the common factor 2 from π. Symmetrizing in k gives Theorem 2 of that note, and $J = \frac{1}{2}$ gives its Theorem 1, $e^{-t/32}[2\cos\frac{\theta}{4}\,\Xi_- - \frac{t}{2}\sin\frac{\theta}{4}\,\Theta_-]/\Xi_+$. Its Theorem 3(a) then follows using the additional spin-$\frac{1}{2}$ matrix symmetry established there: $ED^{1/2}(m) = \lambda D^{1/2}(m_*)$ and $\lambda = E\chi_{1/2}(m)/(2\cos(\theta/4))$ for $0 < \theta < 2\pi$. The cube statement in Theorem 3 also uses independence of its four bridges. Thus every image agrees, while the character identity alone supplies no higher-spin diagonal entries.

4. The centre (Corollary 3)

Proof. Two weights of one representation differ by an element of the root lattice Q, so $e^{2\pi i(1-s)\langle\kappa,H\rangle}$ is independent of κ for every λ iff $\langle\beta, (1-s)H\rangle \in \mathbb{Z}$ for all $\beta \in Q$, i.e. iff $(1-s)H \in Q^* = P^\vee$. Then $e^{2\pi i(1-s)\langle\kappa,H\rangle} = e^{2\pi i\langle\lambda,(1-s)H\rangle}$ is the scalar by which the central element $z_H = e^{2\pi i(1-s)H}$ acts in λ. For $s = p/q$ in lowest terms, $(1-s) = (q-p)/q$ with $\gcd(q-p, q) = 1$, and $(q-p)H/q \in P^\vee$ iff $H/q \in P^\vee$ (Bézout, since $H \in Q^\vee \subset P^\vee$). So the central images are $H \in qP^\vee \cap Q^\vee$, and the elements reached are $e^{2\pi i(q-p)Y}$ with $Y \in P^\vee \cap q^{-1}Q^\vee$: the q-torsion of $P^\vee/Q^\vee \cong Z(G)$, on which multiplication by $q-p$ is an automorphism. For $SU(N)$, $Z(G) = \mathbb{Z}_N$ and its q-torsion is $\mathbb{Z}_{\gcd(q,N)}$. □

For the necessity of the criterion, the adjoint representation of each simple factor contains zero and every root; constancy of its phases forces the root pairings to be integral. For the Bézout step, choose integers a, b with $a(q-p) + bq = 1$ and write $H/q = a(q-p)H/q + bH \in P^\vee$. These give both directions explicitly.

Consequences. Dyadic refinement ($q = 2^n$) reaches the 2-primary part of the centre: all of it for $SU(2)$ at the first halving, ± 1 at a halving and $\mathbb{Z}_4$ at quarters for $SU(4)$, nothing for $SU(3)$ or any odd N. The image classes of a halving, among the groups $SU(N)$, are all central only for $SU(2)$, where $\frac{1}{2}Q^\vee = P^\vee$.

Geometrically the nonidentity central elements in these examples are conjugate cut points of e: the shortest geodesics from e to $e^{2\pi iY}$ ($Y \in P^\vee$ of minimal norm in its class) form the adjoint orbit of Y, a sphere S^2 for $-1 \in SU(2)$ and a $\mathbb{CP}^2$ for $\omega \cdot 1 \in SU(3)$, because conjugation fixes both endpoints. The stabilizers are $U(1) \subset SU(2)$ and $S(U(2) \times U(1)) \subset SU(3)$. The identity has the unique constant minimizing geodesic. The word focal refers here to the degeneracy of the exponential map at the nontrivial central endpoints.

5. $SU(3)$ (Corollary 4)

Use $\mathfrak{t} = \{H \in \mathbb{R}^3 : \sum H_i = 0\}$, $Q^\vee = \mathfrak{t} \cap \mathbb{Z}^3$, $|H|^2 = 2\sum H_i^2$. The weights of $\mathbf{3}$ are $\kappa_i = e_i - \frac{1}{3}(1,1,1)$, so $\langle \kappa_i, H\rangle = H_i$: the image H carries the phase $e^{2\pi i(1-s)H_i}$ on the i-th weight.

- $s = \frac{1}{3}$: the phases $e^{4\pi i H_i/3}$ agree iff $H_1 \equiv H_2 \equiv H_3 \pmod 3$, i.e. $H \in 3P^\vee$. For $(1,1,-2)$ and its Weyl images all $H_i \equiv 1$ and the phase is $e^{4\pi i/3} = \omega^2$; for $(-1,-1,2)$ and its images it is ω. The antifundamental gets the conjugates; the adjoint, with weights $e_i - e_j$, gets $e^{4\pi i(H_i - H_j)/3} = 1$. These images have $|H|^2 = 12$, so $G_H/G_0 = e^{-24\pi^2/t + 2\pi\langle X,H\rangle/t}$. The root images, such as $(1,-1,0)$ with $|H|^2 = 4$ and $G_H/G_0 \approx e^{-8\pi^2/t}$, give the fundamental the phases $\omega^2, \omega, 1$: weight-dependent.
- $s = \frac{1}{2}$: the phases $e^{\pi i H_i}$ agree iff all H_i have the same parity; with $\sum H_i = 0$ they are then all even, $H \in 2Q^\vee$, and the phase is 1.

An $SU(3)$ halving produces identity phases on $2Q^\vee$ and weight-dependent phases otherwise. For a root image $(1,-1,0)$ at halving, the fundamental and antifundamental have phases $(-1,-1,1)$; the adjoint has phases $(-1)^{H_i - H_j}$ on its six roots and 1 on its two zero weights. At trisection the antifundamental phases are the conjugates of the fundamental ones, and the adjoint root phases are $e^{4\pi i(H_i - H_j)/3}$, with two additional 1's. This confirms both cuts on $\mathbf{3}, \overline{\mathbf{3}}, \mathbf{8}$.

The norms follow directly: $2(1+1+4) = 12$ for $(1,1,-2)$ and $2(1+1) = 4$ for $(1,-1,0)$. The exact exponential ratio for any image is $G_H/G_0 = \exp[-(2\pi^2|H|^2 - 2\pi\langle X, H\rangle)/t]$. Consequently the difference $16\pi^2$ between the two displayed exponents is their value at $X = 0$; at general X it includes the linear endpoint terms and the polynomial amplitudes. At singular X one must first combine the images and take the regular limit.

Remark (a trisection as three fusing vortices; semiclassical placement). A trisection inserts two layers, at $s = \frac{1}{3}$ and $\frac{2}{3}$; the marginal law of each is (1). For a central image $H = 3Y$, $Y \in P^\vee$, $z = e^{2\pi iY}$, the image path $e^{iu(X - 2\pi H)}$ passes the two layers at $e^{iX/3}z^{-1}$ and $e^{2iX/3}z^{-2}$, so each of the three sub-segments carries the increment $e^{iX/3}z^{-1}$: every sub-face of the cut face gains the same central flux z^{-1} relative to the direct path, and the three fuse to $z^{-3} = 1$. The $SU(2)$ halving is the two-piece case, where each half-face gains -1 and the two fuse to 1, Dirac's belt. For $SU(3)$ these increments admit a thin-centre-flux interpretation on the three sub-faces, fusing to the identity. The trialities also add to zero for three quarks; selecting a colour singlet additionally requires an invariant tensor. Theorem $1'$ makes the placement exact at the level of character moments: for $s_1 = \frac{1}{3}$, $s_2 = \frac{2}{3}$ and a central image $H = 3Y$ the phase is $e^{2\pi i(2\langle \kappa_1, Y\rangle + \langle \kappa_2, Y\rangle)} = \zeta_{\lambda_1}(z)^2 \zeta_{\lambda_2}(z)$ with $\zeta_\lambda(z) = e^{2\pi i\langle \lambda, Y\rangle}$ the central character, which is the phase of the layers z^{-1} and z^{-2} since $\zeta^3 = 1$. Joint moments of class functions do not determine the full joint law of the two layers, whose relative orientation remains to be described. The central phase in this remark is ACCEPT (A1'). Identifying an image with a physical vortex configuration or a positive path sector requires additional measure information: the amplitudes in (1) and ($1'$) include signed Weyl polynomials. In particular G_H alone gives no probability for such a vortex.

5b. Fundamental trisection and first image shells

For $t > 0$, write $X = \mathrm{diag}(x_1, x_2, x_3)$ with $\sum x_i = 0$ and initially $A_\rho(X) \neq 0$. Put $u = t/3$, $z_{ij} = z_i - z_j$, and $\Delta(Z) = z_{12}z_{13}z_{23}$. The three polynomial prefactors are

$$
\begin{aligned}
P_1(Z) &= z_{23}(z_{12}+iu)(z_{13}+iu),\\
P_2(Z) &= z_{13}(z_{12}-iu)(z_{23}+iu),\\
P_3(Z) &= z_{12}(z_{13}-iu)(z_{23}-iu).
\end{aligned}
\tag{2}
$$

Thus $P_i(Z)=\Delta(Z+iu\kappa_i)$, with the coordinates of κ_i regarded here as ordinary triples. The shifts of differences are what matters; no metric identification is implicit in (2). Define the two explicitly weighted polynomials

$$
\begin{aligned}
F_X(Z) &= \sum_{i=1}^{3} e^{ix_i/3}P_i(Z),\\
F_X^{ab}(Z) &= \omega^2 e^{ix_a/3}P_a(Z)+\omega e^{ix_b/3}P_b(Z)+e^{ix_c/3}P_c(Z),
\end{aligned}
$$

where $a\neq b$ and c is the remaining index. Let $R_{ab}=e_a-e_b$ and $C_a=(1,1,1)-3e_a$. The six R_{ab} are the leading root images and the two triples $\{C_a\}$, $\{-C_a\}$ are the leading nonzero central images. Formula (1) becomes the exact identity

$$
E\chi_{\mathbf{3}}(m)=e^{-t/27}\frac{\mathcal{N}_0+\mathcal{N}_R+\mathcal{N}_C+\mathcal{R}_N}{\mathcal{D}_0+\mathcal{D}_R+\mathcal{D}_C+\mathcal{R}_D},\qquad s=\frac{1}{3},
\tag{3}
$$

with the following terms displayed without a small-t expansion:

$$
\begin{aligned}
\mathcal{N}_0 &= F_X(X), & \mathcal{D}_0 &= \Delta(X),\\
\mathcal{N}_R &= e^{-8\pi^2/t}\sum_{a\neq b} e^{4\pi(x_a-x_b)/t}F_X^{ab}(X-2\pi R_{ab}),\\
\mathcal{D}_R &= e^{-8\pi^2/t}\sum_{a\neq b} e^{4\pi(x_a-x_b)/t}\Delta(X-2\pi R_{ab}),\\
\mathcal{N}_C &= e^{-24\pi^2/t}\sum_{a=1}^{3}\Big[\omega^2 e^{-12\pi x_a/t}F_X(X-2\pi C_a)\\
&\qquad\qquad + \omega e^{12\pi x_a/t}F_X(X+2\pi C_a)\Big],\\
\mathcal{D}_C &= e^{-24\pi^2/t}\sum_{a=1}^{3}\Big[e^{-12\pi x_a/t}\Delta(X-2\pi C_a)\\
&\qquad\qquad + e^{12\pi x_a/t}\Delta(X+2\pi C_a)\Big].
\end{aligned}
\tag{4}
$$

For precision the remainders are exact sums. With $\mathcal{T}=Q^\vee\setminus(\{0\}\cup\{R_{ab}\}\cup\{\pm C_a\})$, $Z_H=X-2\pi H$ and $r_H=\exp[-2\pi^2|H|^2/t+4\pi\sum_i x_iH_i/t]$,

$$
\begin{aligned}
\mathcal{R}_N &= \sum_{H\in\mathcal{T}} r_H\sum_i e^{ix_i/3}\omega^{2H_i}P_i(Z_H),\\
\mathcal{R}_D &= \sum_{H\in\mathcal{T}} r_H\Delta(Z_H).
\end{aligned}
\tag{5}
$$

Written derivation. In the dual metric $|\kappa_i|^2=1/3$, giving $ts(1-s)|\kappa_i|^2/2=t/27$. For a covector v, $\pi(v)=\frac{1}{2}(v_1-v_2)(v_1-v_3)(v_2-v_3)$, so

$$
\pi(iZ^\flat/t-2\kappa_i/3)=(-4i/t^3)P_i(Z),\qquad \pi(iZ^\flat/t)=(-4i/t^3)\Delta(Z).
$$

Cancel $(-4i/t^3)G_0$ from numerator and denominator of (1). For R_{ab}, the three phases are $\omega^2,\omega,1$; for C_a they are all ω^2 and for $-C_a$ all ω. Also $\sum_i x_i(C_a)_i=-3x_a$. These substitutions prove (3)–(5). In particular the exact zero-image sector is

$$e^{-t/27}\sum_i e^{ix_i/3}\prod_{j\neq i}\left(1+\frac{it}{3(x_i-x_j)}\right). \tag{6}$$

The full expression extends continuously across affine walls. The remainders start at $|H|^2 = 16$, hence at Gaussian exponent $32\pi^2$ at $X = 0$: ordering an integral zero-sum triple shows the only nonzero norms below 16 are 4 and 12, represented by the two displayed shells. At a general endpoint the linear terms in r_H remain essential. Discarding (5) is a formal image truncation until a normalized bound on the chosen endpoint set has been proved. In particular the polynomial zeros at $X = 0$ require the complete regular limit.

6. What the result says, and what it leaves open

A5: ACCEPT the threshold; REFINE the inference from it. The result is an exact statement about one bridge, the building block of the parallel insertion, for every compact simply connected group, every representation and every cut fraction. It concerns characters only: the diagonal entries of the midpoint matrix $E\,D^\lambda(m)$ differ from the weight terms of (1), since Theorem 4 of the $SU(2)$ note gives the spin-1 entries an additional Gaussian tail, so atlas cell 1 is unaffected. In the refinement it fixes which relative Cartan images of a single step have central phases. The exponents at the identity, $8\pi^2$ (roots) and $24\pi^2$ (central $SU(3)$ trisection images), exceed the formal per-volume threshold of the zero-spacing note. Indeed, with $t = g^2_{\mathrm{lat}}$ and one-loop running $t^{-1} = 2b_0\log(1/(a\Lambda))$, $b_0 = 11/(16\pi^2)$, a per-cell bound $Ct^{-r}e^{-c/t}$ would sum in a fixed four-volume to $O((\log(1/a))^r a^{2b_0c-4})$. A strict positive power requires $c > 2/b_0 = 32\pi^2/11$; the two displayed exponents give powers $2b_0c = 11$ and 33. Logarithmic corrections matter at equality. The bridge formula supplies neither that normalized per-cell bound uniformly in boundary data nor its stability under repeated blocking. The endpoint term $2\pi\langle X, H\rangle/t$, affine-wall limits and polynomial prefactors must be controlled first. Character moments also leave the relative orientations needed by Ψ undetermined.

Centre-valued link configurations exist on any lattice whatever the blocking, and thick centre vortices (Mack and Petkova 1979; 't Hooft 1978; metadata) are a question about many steps. In 't Hooft's language (2026-09-29, from memory, references Crossref-verified): multiplying one mid-edge by a central element inserts his disorder operator on the dual set of that edge's faces, so the open question of whether refinement ties to centre vortices asks whether image terms at a cut p/q generate such insertions; on a finite torus his twisted boundary conditions and electric and magnetic flux sectors ('t Hooft 1979; metadata) organize the cycle holonomies whose small-angle part is the non-based cycle phase of round 12 in the small-field note §10; and the fractional topological charge $1/N$ mod 1 of twisted configurations ('t Hooft 1981; metadata; value recalled) is $

rac13$ for $SU(3)$, the group whose centre only trisections reach. Pure $SU(3)$ has exact centre symmetry and no dynamical fundamental matter to screen its triality charges; whether that symmetry is broken depends on the regime. Fradkin and Shenker (1979) (abstract, checked 2026-09-28) establish an analytic connection between confinement and Higgs regions for fundamental lattice Higgs matter. Osterwalder and Seiler (1978) (metadata) supplies related gauge-field background. Applying Higgs complementarity directly to fermionic QCD would exceed these statements. Dynamical fundamental quarks can screen triality and explicitly break the corresponding centre symmetry; the algebraic phases of (1) remain properties of the pure heat-kernel bridge. These distinctions belong in the joint-paper comparison. Whether a triadic refinement, $b = 3$, organizes the $SU(3)$ large-field terms better than dyadic refinement is open; factor-2 decimation with the $\mathbb{Z}_2$ factor kept explicit goes back to Tomboulis (1981) (metadata).

Blocking and centre prior art (Round 5B B2, bounded search, 2026-09-28). The following DOI/title pairs were verified through Crossref; the reading labels specify the evidence used.

- Gupta, Guralnik, Patel, Warnock and Zemach (1984), *Monte Carlo Renormalization Group for SU(3) Lattice Gauge Theory* (abstract): SU(3) MCRG estimates both the beta function and the generated effective action. This supplies direct SU(3) precedent for the blocking programme, with no centre-based choice of factor claimed in the abstract read here.
- Bhattacharya, Gupta and Lee (2000), *Fixed-point pure gauge action using $b = \sqrt{3}$ RGT* (passage, §1, eqs. (1)–(4)): the block basis uses cube body diagonals, and two successive transformations satisfy $T_2T_1 = 3I$. This is a concrete net factor-3 construction through two rotated $\sqrt{3}$ steps. Its stated motivations concern geometry, locality and degrees of freedom. The coordinate twists of its intermediate lattice are specified by eq. (2); those coordinate identifications alone supply no $\mathbb{Z}_3$ vortex twist.
- Tomboulis (2007), *Deriving confinement via RG decimations* (passage, §§2–4 and conclusion): potential-moving decimations with scale factor b are applied to partition functions with and without external centre flux. The explicit group is SU(2); SU(3) is proposed as an extension in the conclusion. This supplies a centre-flux observable and a decimation strategy to compare with Ψ. Its confinement claim is recorded as the author's proposal, with no proof of that claim imported here.
- Kovács–Tomboulis (1999), *SU(3) string tension and the presence of vortices* (abstract): simulations report that centre fluctuations reproduce the heavy-quark potential and persist under smoothing. This concerns thick centre vortices; the abstract supplies no preferred blocking factor.

The search covered SU(3) Migdal–Kadanoff with $b = 3$, MCRG with $b = 3$ or $\sqrt{3}$, and centre-vortex free energies under decimation. Within the passages checked, an explicit SU(3) $b = 3$ Migdal–Kadanoff calculation tied to triality, or a $\mathbb{Z}_3$ vortex-free-energy decimation justified by the centre order, was not located. This is a bounded search outcome. The positive finding is the established $\sqrt{3}$ geometry and its net factor 3, alongside the external-flux decimation strategy.

Relation to Corollary 3 (inference). Equal subdivision into b pieces gives first cut $s = 1/b$ and hence central image phases in $Z(G)[b]$. A general lattice RG transformation also integrates transverse variables and can use rotated blocks, so its length factor alone does not specify this bridge cut. Testing a centre advantage requires an actual map retaining the joint layers and comparing normalized twisted and untwisted observables. Equations (3)–(5) provide explicit character inputs for that test; their image exponents alone leave the estimate open.

7. Consequence for STATE

Atlas §1b: the bullet on which part of the centre a cut reaches now rests on Corollary 3 of an exact formula, (1), which also carries the $SU(2)$ belt-trick phase and the $SU(3)$ triality at thirds. The Round 5B Part A referee is complete, with verdicts and corrections above. The exact trisection formula for the fundamental of $SU(3)$ is now displayed in §5b, including polynomial prefactors and exact residual sums (Part B, written derivation, no independent referee yet). The bounded prior-art search in §6 locates factor-$\sqrt{3}$ and net factor-3 blocking; the centre-adapted normalized refinement estimate remains the next question.